

# SIMATS ENGINEERING

## ASSESSMENT TOOL II — Visualization Analysis Report

|                                         |                  |                                       |
|-----------------------------------------|------------------|---------------------------------------|
| Course: Data Handling and Visualization | Code: DSA06      | CO: CO5                               |
| Duration: 180 min                       | Total Marks: 100 | Reg No : 192324284<br>Name: V.P Sneha |

### COURSE OUTCOME COVERED

CO5: Analyze and evaluate visualization choices to communicate patterns, comparisons, relationships, and potential misleading effects through appropriate visualization methods. (BL4)

### Activity Tool : Visualization Analysis Report

Weightage: 20%

### Code of Conduct:

I V.P Sneha (192324284) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.P Sneha

### Assessment Criteria

| Criteria | Visualization Selection and Understanding 25% | Visualization Analysis and Interpretation 25% | Application of Visualization Principles 20% | Organization and Presentation 20% | Accuracy 10% | Clarity 5% | Total |
|----------|-----------------------------------------------|-----------------------------------------------|---------------------------------------------|-----------------------------------|--------------|------------|-------|
| Q1       |                                               |                                               |                                             |                                   |              |            |       |
| Q2       |                                               |                                               |                                             |                                   |              |            |       |
| Q3       |                                               |                                               |                                             |                                   |              |            |       |
| Q4       |                                               |                                               |                                             |                                   |              |            |       |
| Q5       |                                               |                                               |                                             |                                   |              |            |       |

## Visualization Analysis Report – Assessment Tool

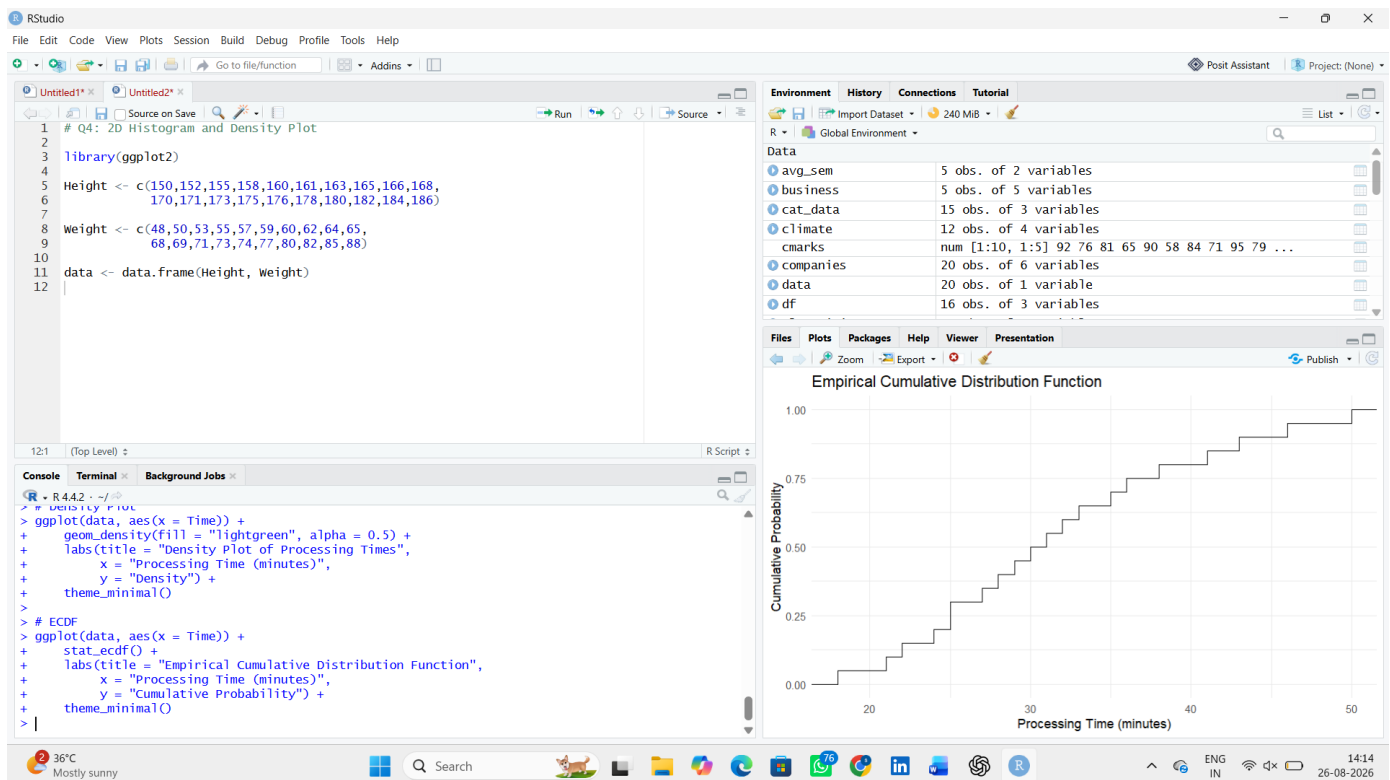
### R Programming Practical Questions with Datasets

Instructions: Answer all five questions. Use R programming to construct the required visualizations, analyze the patterns, compare visualization choices where requested, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

#### Q1. Comparing Distributions – Histogram, Density and ECDF

A quality-control team records the following processing times (in minutes) for 20 randomly selected tasks. Using R, create a histogram and a density plot for the data. Also construct an empirical cumulative distribution function (ECDF). Compare the three visualizations and explain which aspects of the distribution—center, spread, skewness, and cumulative behavior—are communicated most clearly by each.

Data: Time = 18, 21, 22, 24, 25, 25, 27, 28, 29, 30, 31, 32, 33, 35, 36, 38, 41, 43, 46, 50  
*Expected R tasks: histogram, density estimation, ECDF, appropriate labels/scales, comparison and interpretation.*

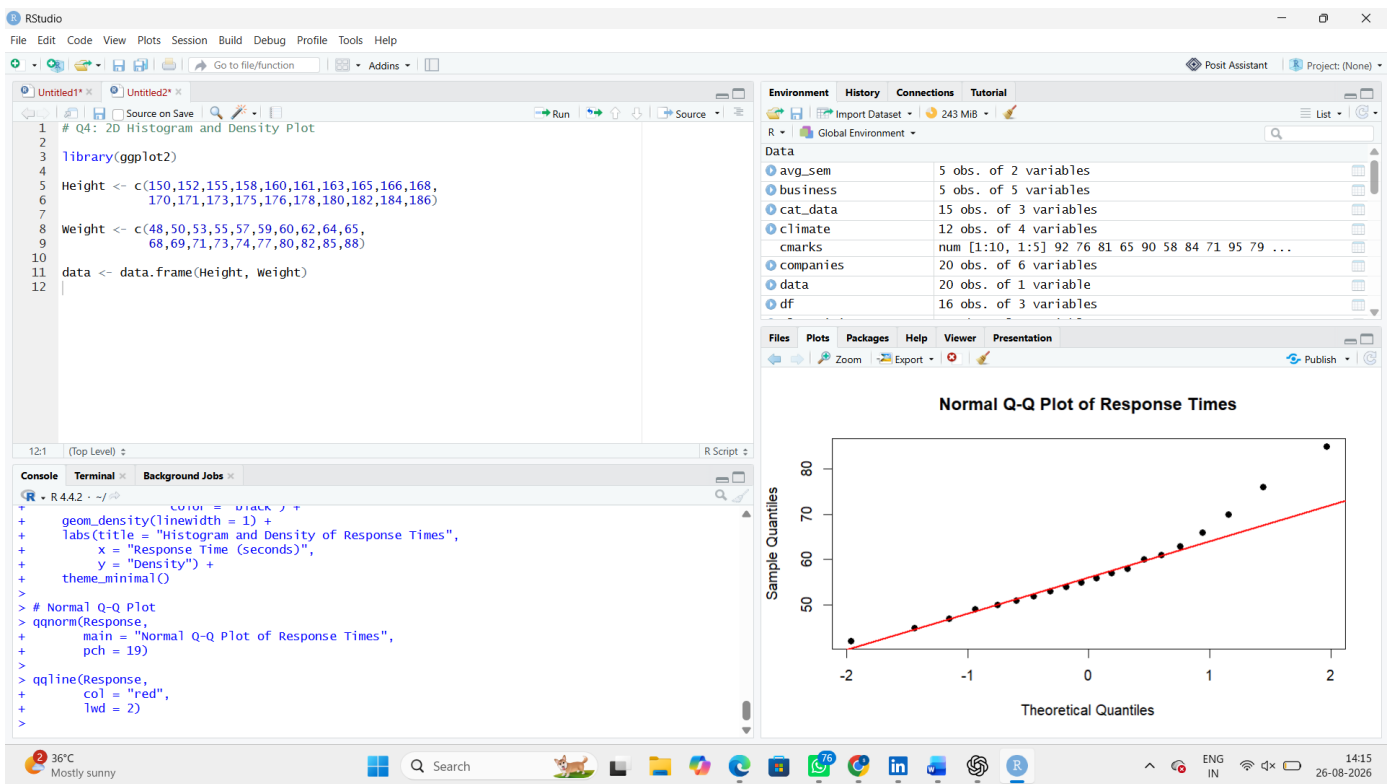


#### Q2. Q-Q Plot and Normality Analysis

A researcher measures the following response times (in seconds). Use R to create a normal Q-Q plot and a histogram/density visualization. Assess whether the observations are reasonably consistent with a normal distribution. Identify any departures from the reference line and explain what they indicate about the distribution.

Data: Response = 42, 45, 47, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 60, 61, 63, 66, 70, 76, 85

*Expected R tasks: qqnorm()/qqline() or ggplot2 Q-Q plot, histogram/density plot, interpretation of deviations and outliers.*

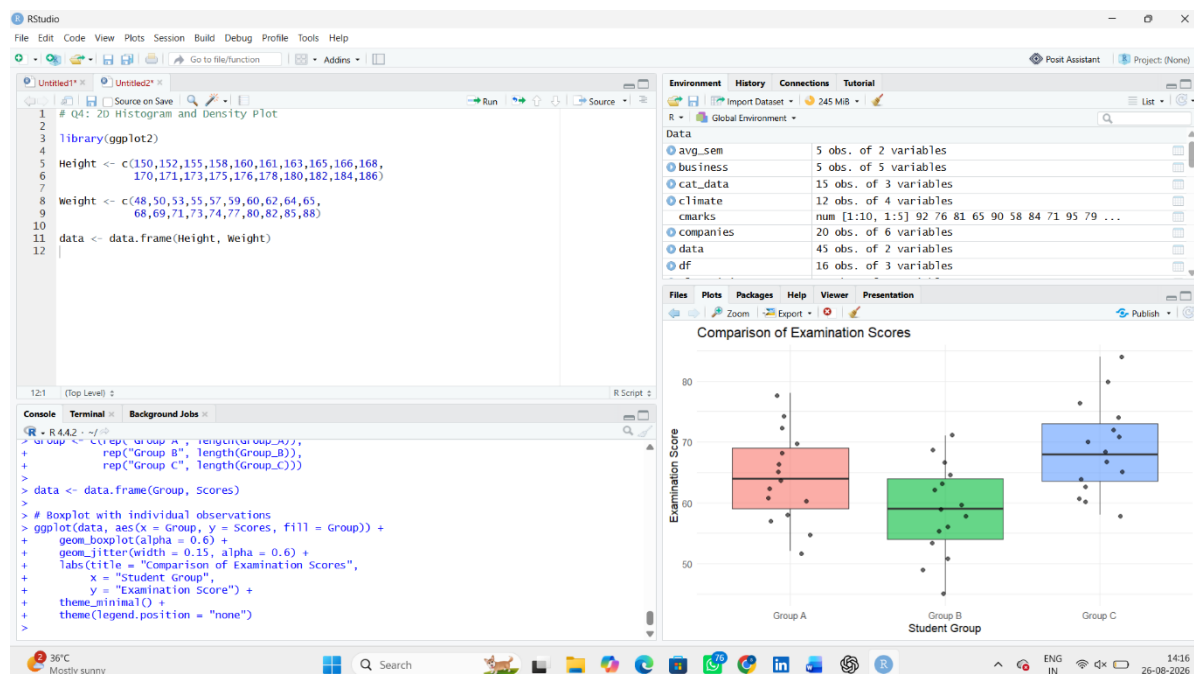


### Q3. Visualizing Many Distributions at Once

Three groups of students have examination scores recorded below. Create a suitable visualization that compares the distributions of all three groups at once. Use boxplots and/or violin plots, and optionally overlay individual observations. Analyze differences in center, spread, and possible outliers. Explain why the selected visualization is more informative than displaying three separate plots.

Group A = 52,55,57,58,60,61,62,64,65,66,68,70,72,74,78;  
 Group B = 45,49,51,53,55,56,58,59,60,62,63,65,67,69,71;  
 Group C = 58,60,61,63,64,65,67,68,70,71,72,74,76,80,84

*Expected R tasks: data reshaping, grouped boxplot/violin plot, optional jitter, comparison of distributions, interpretation.*



## Q4. Relationships and Multivariate Visualization

A study records advertising expenditure, website visits, and sales for 15 observations. Create a scatterplot of advertising expenditure versus sales and use an appropriate visual encoding to incorporate website visits. Add a fitted trend line. Analyze the strength and direction of the relationship and explain whether the additional variable improves the interpretation or risks cluttering the visualization.

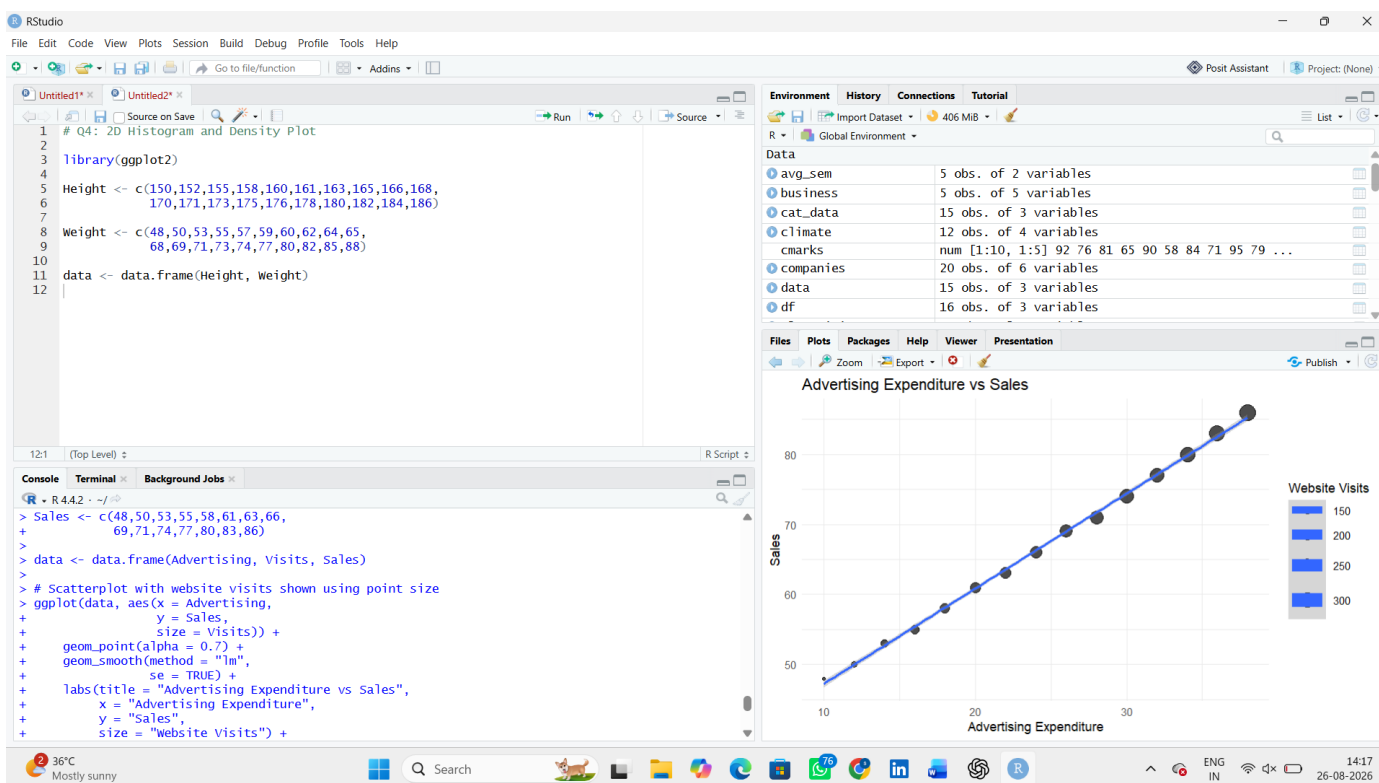
Advertising = 10,12,14,16,18,20,22,24,26,28,30,32,34,36,38;

Visits = 120,135,150,165,180,195,210,225,245,260,275,290,305,320,340;

Sales = 48,50,53,55,58,61,63,66,69,71,74,77,80,83,86

### AT2 – Visualization Analysis Report | Data Handling and Visualization | DSA06

*Expected R tasks: scatterplot, trend line, third-variable encoding, appropriate legend/scale, relationship analysis and critique.*



## Q5. Misleading Visualization and Redesign

A college reports the following pass percentages for seven departments. Create two visualizations: (1) a misleading version using a truncated y-axis and an inappropriate visual emphasis, and (2) a redesigned version using an appropriate baseline, ordered categories, and clear labels. Compare the two plots and explain how the first visualization can exaggerate differences and why the redesigned visualization communicates the data more honestly.

Department = CSE, ECE, EEE, MECH, CIVIL, AIDS, IT;

Pass\_Percentage = 91,93,92,89,90,95,94

*Expected R tasks: bar chart with controlled y-axis, redesigned bar chart, ordering, labels, misleading-axis critique, accessibility and communication analysis.*

